

Full-universe evaluation — $t = 2016$ (all $M[t]=0$ pairs, no sampling)

Method	PR-AUC	AUROC	F1	NDCG@20	Prec@20	CWR
RCA Persistence	0.247	0.652	0.200	0.136	0.137	0.335
Density	0.060	0.782	0.114	0.132	0.122	0.886
ECI	0.019	0.469	0.036	—	—	0.426
KNN (LLM embeddings)	0.035	0.642	0.073	0.087	0.078	0.678
XGBoost (unoptimized)	0.199	0.895	0.292	0.324	0.289	0.948
XGBoost (optimized)	0.223	0.901	0.309	0.340	0.309	0.955
GNN-4F	0.076	0.819	0.141	0.110	0.101	0.912
GNN-11F	0.081	0.825	0.148	0.137	0.121	0.919
GNN-11F+LLM	0.082	0.827	0.150	0.159	0.141	0.921
GNN-11F-LLM + RCA-skip (unoptimized)	0.196	0.893	0.299	0.288	0.266	0.946
GNN-11F-LLM + RCA-skip (optimized)	0.204	0.893	0.300	0.289	0.263	0.948

Bold row = best method overall (wins the most metrics: XGBoost (optimized)). ECI has no within-country ranking signal, so NDCG@20 / Prec@20 are marked — rather than reported off an arbitrary tie-break.